

Elektrik (Net)

a) Vektörler

$$\rightarrow \text{Kuvvet } (\vec{F}) = \frac{q_1 q_2}{d^2} \cdot \vec{k}$$

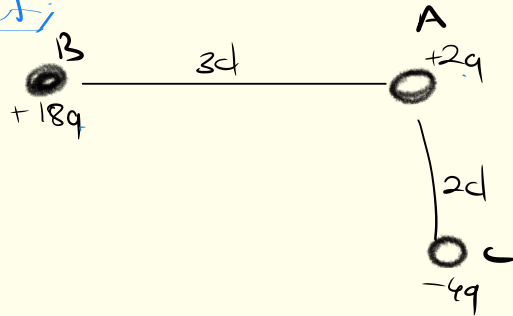
$$\rightarrow \text{Elektrik Alan } (\vec{E}) = \frac{q \cdot \vec{k}}{d^2}$$

b) Skalerler

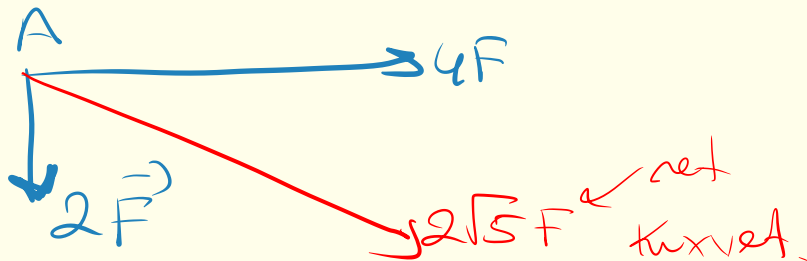
$$\rightarrow \frac{q_1 q_2 \cdot k}{d} = \text{Elektriksel potansiyel Enerji: } (U)$$

$$\rightarrow \frac{q \cdot k}{d} = \text{Elektriksel potansiyel } q \text{ da potansiyel fark } (\checkmark)$$

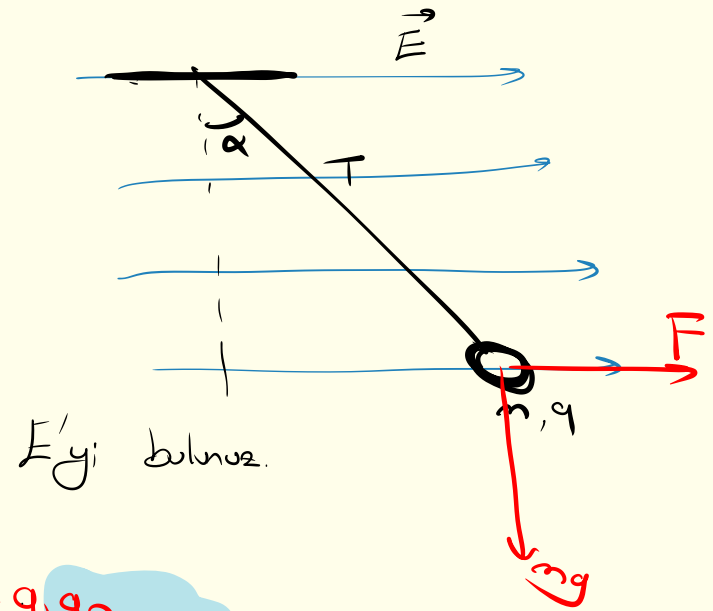
1) Kuvvet;



Soru; A cismi üzerindeki net kuvveti bulun.



2. Tip



E_y bulunuz.

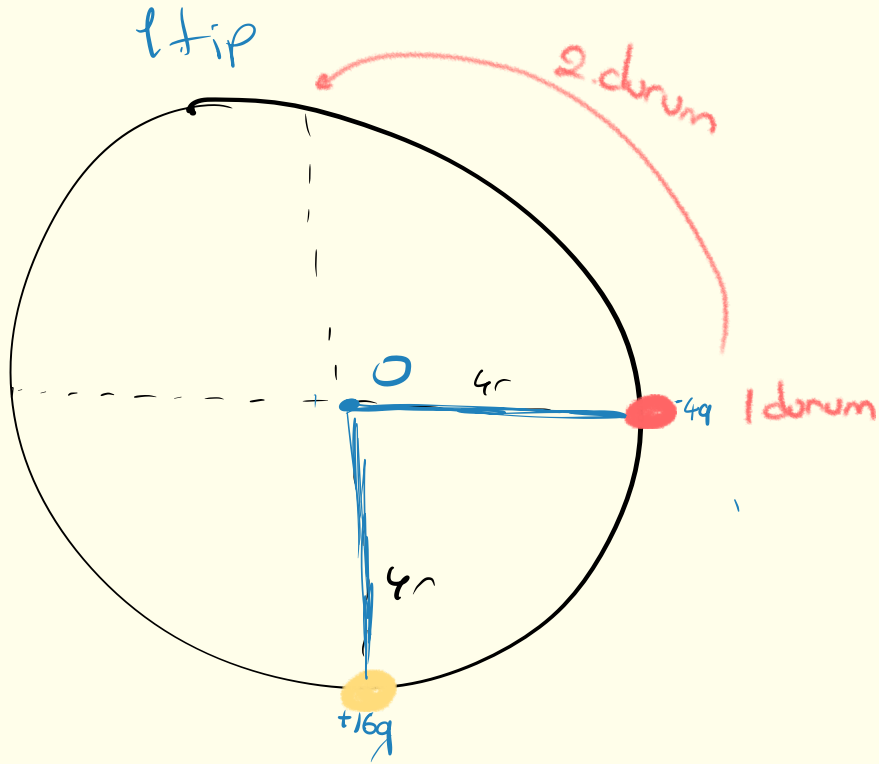
$$F = \frac{q_1 q_2}{d^2} \cdot k$$

$$F = q \cdot E$$

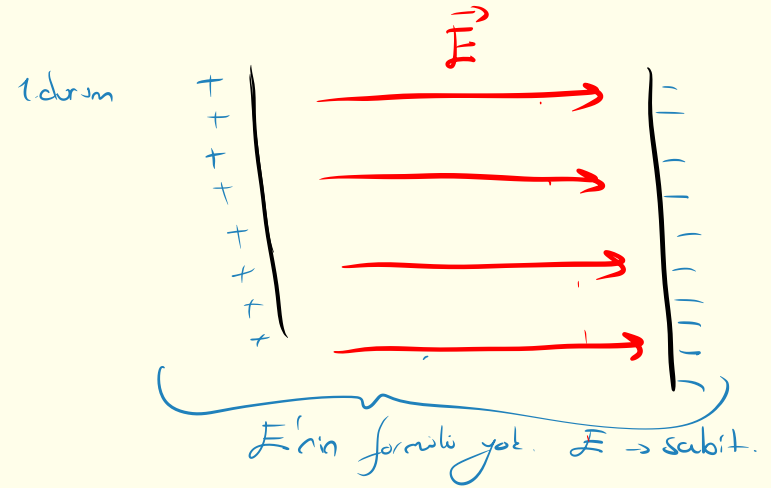
$$\tan \alpha = \frac{F}{mg} = \frac{Eq}{mg} \Rightarrow \boxed{E = \frac{\tan \alpha \cdot mg}{q}}$$

Elektrik Alanı,

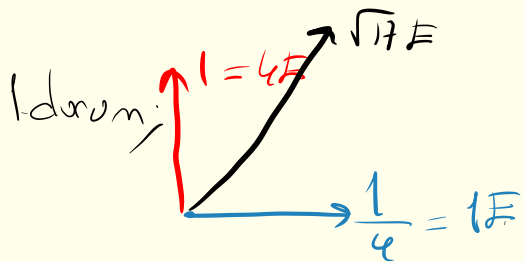
Bir yükten +1'e yaptığı kuvettir.



2. Tip (Paralel levhalar: elektrik alanı)



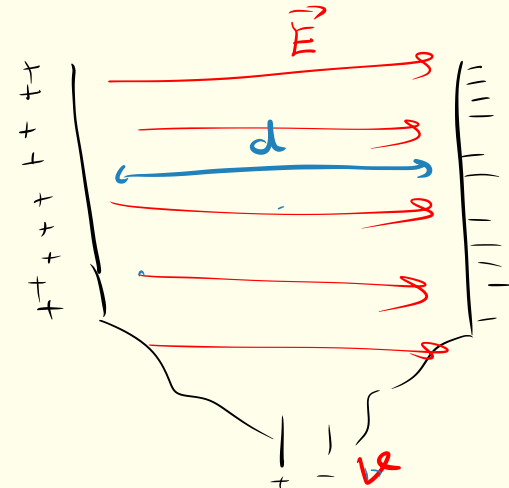
2. durumda da toplam Elektrik Alanı bulunur;



2. durum

$$\frac{4q \cdot k}{16r^2} = \frac{kq}{4r^2} = E \quad \uparrow 4E$$

$$\frac{16q \cdot k}{16r^2} = \frac{kq}{r^2} = 4E \quad \uparrow 1E$$

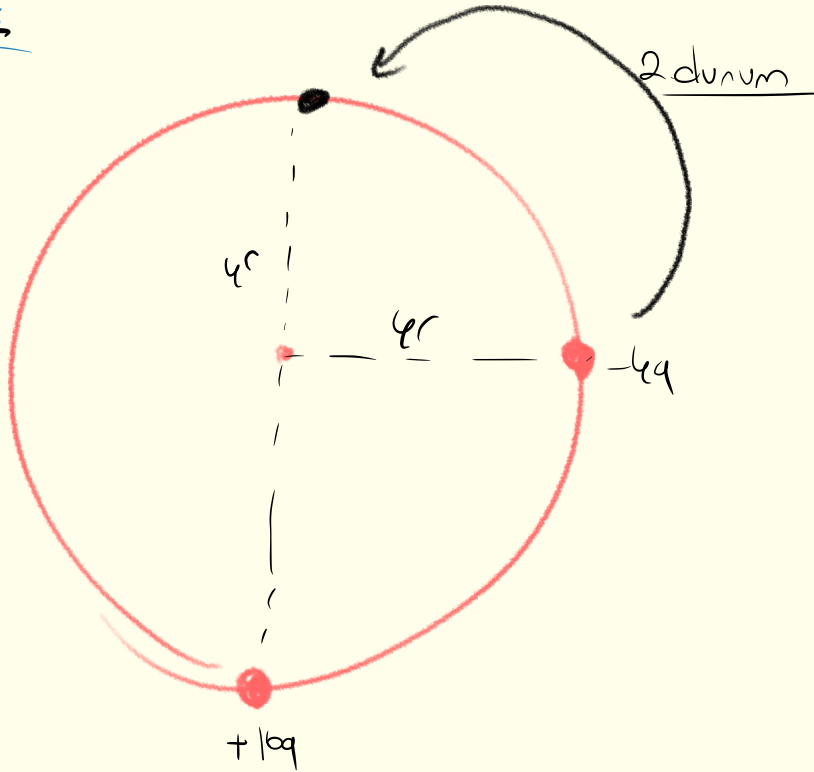


E'nin formülü var.

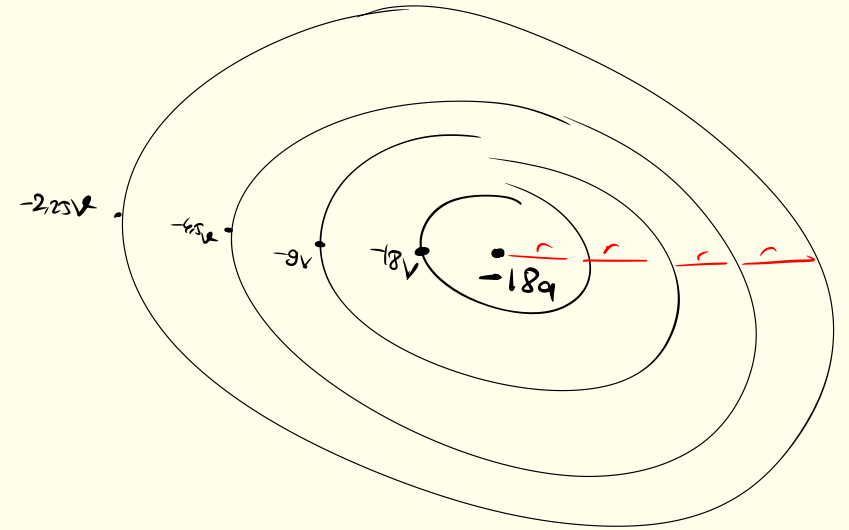
$$E = \frac{V}{d}$$

Elektriksel potansiyel
ya da
Elektriksel potansiyel fark $\Rightarrow V = \frac{q \cdot k}{d}$

Örnek 1



Örnek 2



1. ve 2. durumlarda V 'leri
bulun

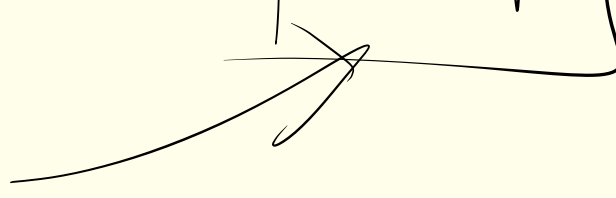
$$\frac{k \cdot 91}{d} \quad \frac{k \cdot 16q}{4r} = \frac{k \cdot 4q}{r}$$

$$\frac{k \cdot -4q}{4r} = -\frac{k \cdot q}{r}$$

$$\boxed{V = -\frac{3kq}{r}}$$

2. Durum

μ aynı

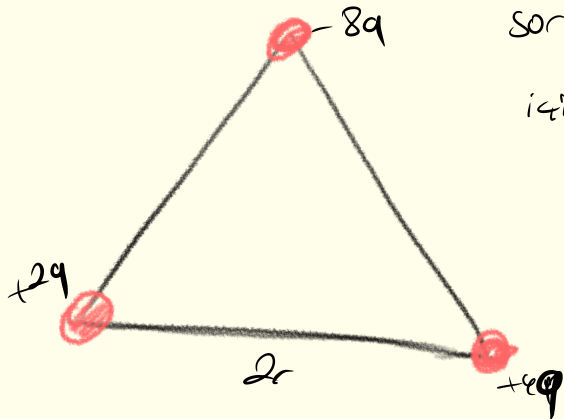


Elektriksel potansiyel Enerji hesaplaması $\Rightarrow \Delta E = \text{yapılan iş}$

$$E_{\text{son}} - E_{\text{ilk}} = W$$

$$k \frac{q_1 q_2}{d} = U$$

1. Tip



soru) $2q$ 'yı sonuca götürmek için yapılan işi bulun.

$$\frac{-8q \cdot +4q \cdot k}{2r} = \frac{-16q^2 k}{2r} \downarrow E_{\text{son}}$$

$$\frac{-8q \cdot +2q \cdot k}{2r} = \frac{-8q^2 k}{2r} \quad \frac{-8q \cdot +4q \cdot k}{2r} = \frac{-16q^2 k}{2r} \quad \frac{+2q \cdot +4q \cdot k}{2r} = \frac{+8q^2 k}{2r}$$

E_{ilk}

$$W = \frac{-16q^2 k}{2r} + \frac{+8q^2 k}{2r} = \frac{-8q^2 k}{2r}$$

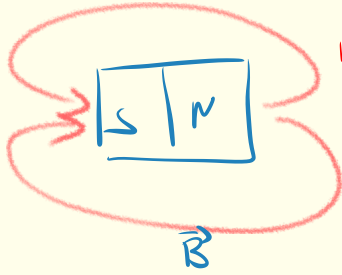
$$W > 0$$

$\rightarrow$ Elektriksel kuvvetler iş yapmışlar.

Manyetizama

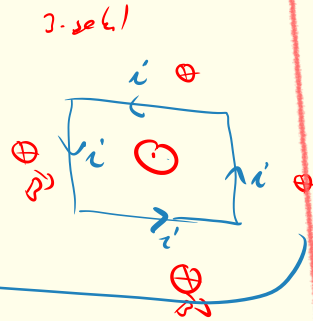
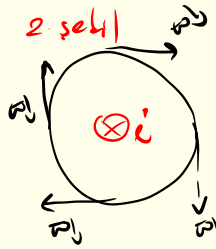
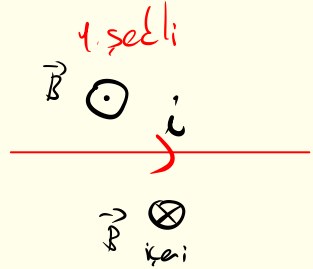
Soru Manyetik Alan nasıl oluşur?

→ Mikrotis;



Nereye bağlı; Mikrotis ve uzaklığa bağlı.

→ üzerinden Akım geçen tel;

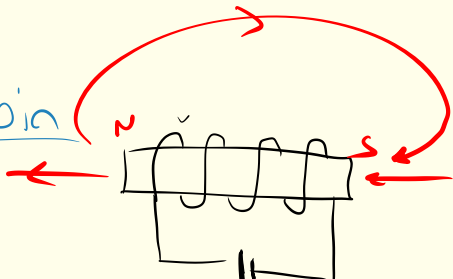


$$B = \frac{2 \mu_0 i}{d}$$

eğer tel çember ise

$$B = \frac{\mu_0 i}{d}$$

3. Bobin

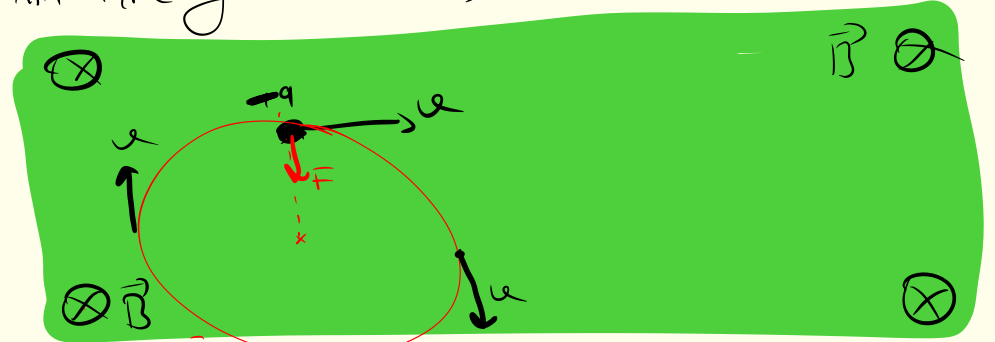


$$\frac{4 \mu_0 i N}{d} \rightarrow \text{sum sayıları}$$

2. soru

Ne yapar?

$\vec{B}$ 'nin işte yük atarsak; (çembersel hareket yapar)



4 parmak $\vec{B}$
parmak $\vec{v}$
Aynı içi kuvvet $\vec{F}$ (+ ise)
Aynı dışı (- ise)

$$F_{\text{net}} = F_{\text{cent}} \\ F = B \cdot q \cdot v = \frac{mv^2}{r}$$

$$Bq = \frac{mv}{r}$$

$$F_{\text{magnetik}} = B \cdot q \cdot v$$

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